Week-2 Solution Sheet

1. What are “Black Swans”?
    a. Frequent events with predictable and minor impacts.
    b. Outlier events that often have extreme and unpredictable impacts.
    c. Known risks that are intentionally ignored during model training.
    d. Events that follow a thin-tailed distribution and have negligible consequences.

    Explanation: Robustness Part 1 [3:36-5:12] - Black swans are events that are outliers, lying outside typical expectations, and often carry extreme impact.

2. Which of the following is considered a “Known Unknown”? **(Select all that apply.)**
    a. Recollection
    b. Closed-ended Questions
    c. Meta-ignorance
    d. Conscious Ignorance
    e. Open-ended Exploration
    f. Self-analysis

    Explanation: Robustness Part 1 [19:20-22:05] - “Known Unknowns” are things we are aware of but don’t understand such as known classic risks/conscious ignorance or closed-ended questions.

3. Which type of distribution shift occurs when the input distribution $P(x)$ changes between training and testing but $P(y|x)$ remains constant?
    a. Concept shift
    b. Normalized shift
    c. Covariate shift
    d. Prior probability shift

    Explanation: Robustness Part 2 [23:00-24:53] - P(x) changes between train and test, but p(y|x) does not change in covariate shift.

4. What is “transferability” in the context of adversarial attacks?
    a. The ability to move a model from a secure state to a vulnerable state.
    b. The phenomenon where adversarial examples crafted for one model also fool another model.
    c. Converting a black-box model into a white-box model.
    d. The process of retraining a model on adversarial attacks.

    Explanation: Lecture 8 and 9 Lecture Notes (Slide 70): Adversarial examples crafted for one model, can be used to attack other models is known as transferability.

5. According to the lecture, what is a common trade-off when using Adversarial Training to improve robustness?
    a. It requires a significantly larger dataset than normal training.
    b. It can significantly reduce the accuracy on clean examples.
    c. It only works for simple binary classifiers.
    d. It requires a low-variance dataset to function effectively.

Explanation: Lecture 8 and 9 Lecture Notes (Slide 78): Currently, adversarial training can reduce accuracy on clean examples by 10%+.

6. Which of the following are challenges associated with direct human ratings when modeling preferences? **(Select all that apply.)**
    a. Human-in-the-loop evaluation is expensive and time-consuming.
    b. Direct ratings require white-box access to the model.
    c. Human ratings are always completely objective and free from annotator bias.
    d. Human judgments are noisy and miscalibrated.

Explanation: RLHF [8:39-13:31] - There are two problems with human ratings: human-in-the-loop is expensive and human judgments are noisy and miscalibrated.

7. What does Direct Preference Optimization (DPO) represent in the RLHF framework?
    a. RLHF without the reward model
    b. RLHF without human feedback
    c. RLHF without reinforcement learning
    d. RLHF without policy optimization

Explanation: AI Alignment [4:54-6:35] - RLHF – Reward model = Direct Preference Optimization (DPO)

8. What is the first step of the RLHF pipeline?
    a. Collecting comparison data and training a reward model.
    b. Optimizing a policy against the reward model using reinforcement learning.
    c. Collecting demonstration data and training a supervised policy.
    d. Performing external white-box auditing of the pretrained model.

Explanation: RLHF [3:57-6:38] - The first step in the RLHF pipeline is to collect demonstration data and train a supervised model.

9. Why is black-box access insufficient for rigorous AI audits?
    a. It is too expensive for external auditors.
    b. It does not allow for human-in-the-loop feedback.
    c. It prevents the use of reinforcement learning.
    d. It only allows auditors to query the system and observe outputs.

Explanation: AI Alignment [24:28-25:13] - Auditors can only query the system and observe its output.

10. According to the lecture, what does the Kullback–Leibler (KL) divergence penalty prevent in RLHF?
    a. The reward model from becoming too complex
    b. The policy model from diverging too far from the initial pretrained model
    c. Human evaluators from providing biased feedback
    d. The model from over-generalizing

Explanation: RLHF [14:31-16:35] - Kullback–Leibler (KL) divergence is the penalty which prevents us from diverging too far from the pretrained model.

11. Which of the following is true? **(Select all that apply.)**
    a. Extremistan has thin tails while Medicristan has long tails.
    b. Extremistan is determined by many small events.
    c. Mediocristan is the tyranny of the collective.
    d. Extremistan's impact is potentially scalable.

    Explanation: Robustness Part 1 [17:22-19:19]

12. Identify the equations that can lead to a long-tailed distribution.
    a. Idea * student * resources * time
    b. Idea * student + resources * time
    c. Idea + student + resource + time
    d. Idea - student * resource - time

    Explanation: Robustness Part 1 [15:17-17:21] - Research output = Ideas X Time X Students X Resources